

Índice

Math

matrizFibo

```
1 struct Matrix {
2     long long mat[2][2];
3     Matrix friend operator *(const Matrix &a, const
4         Matrix &b){
5         Matrix c;
6         for (int i = 0; i < 2; i++) {
7             for (int j = 0; j < 2; j++) {
8                 c.mat[i][j] = 0;
9                 for (int k = 0; k < 2; k++) {
10                     c.mat[i][j] += a.mat[i][k] * b.mat
11                         [k][j];
12                 }
13             }
14         }
15     };
16 Matrix matpow(Matrix base, long long n) {
17     Matrix ans{ {
18         {1, 0},
19         {0, 1}
20     } };
21     while (n) {
22         if(n&1)
23             ans = ans*base;
24         base = base*base;
25         n >>= 1;
26     }
27     return ans;
28 }
29 ll fib(int n) {
30     Matrix base{ {
31         {1, 1},
32         {1, 0}
33     } };
34     return matpow(base, n).mat[0][1];
35 }
```

Miscelanea

Plantilla

```
1 #include <bits/stdc++.h>
2 #include <bits/extc++.h>
3 using namespace std;
4 using namespace __gnu_pbds;
5 typedef long long ll;
6 typedef long double ld;
7
8 typedef tree<pair<ll, ll>, null_type, less<pair<ll,
9     ll>>, rb_tree_tag,
10     tree_order_statistics_node_update>
11     ordered_set_men;
12 typedef tree<int, null_type, greater<int>,
13     rb_tree_tag, tree_order_statistics_node_update>
14     ordered_set_may;
15
16 #define CRISTIANO RONALDO GANO 35 COPAS \
17     ios_base::sync_with_stdio(false); \
18     cin.tie(NULL); \
19     cout.tie(nullptr);
20
21 #define hola cout << "hola" << endl;
22 #define YES cout << "YES" << endl;
23 #define NO cout << "NO" << endl;
24 #define dbg(x) cout << #x << ": " << x << endl;
25 #define dbg2(x, y) cout << #x << ": " << x << " __ "
26     << #y << ": " << y << endl;
27
28 #define printvii(v_v) \
29     for (auto [x_x, y_y] : v_v) \
30     { \
31         cout << x_x << " " << y_y << endl; \
32     } \
33     cout << endl;
34
35 #define printst(st) \
36     for (auto num : st) \
37     cout << num << " "; \
38     cout << endl;
39
40 template <typename T>
41 void printv(T v)
42 {
43     for (auto x : v)
```

```

35         cout << x << " ";
36     cout << endl;
37 }
38
39 #define RAYA cout << "-----"
40     << endl;
41 // #define F first
42 // #define S second
43 const ll MOD = 1'000'000'007;
44 const vector<int> F = {0, 1, 0, -1};
45 const vector<int> C = {1, 0, -1, 0};
46
47 int main()
48 {
49     CRISTIANO_RONALDO_GANO_35_COPAS
50     //sol
51     return 0;
52 }

```

Datastructures

LowestCommonAncestor

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5 int n, m; //nodos, links
6 vector<int> h; //altura del nodo
7 vector<int> primera; //primera aparicion
8 // set<int> st;
9 vector<bool> vis;
10 vector<vector<int>> gf;
11 vector<int> nodo; //nodo
12
13 void dfs(int currt, int alt)
14 {
15     if (!vis[currt])
16     {
17         primera[currt] = h.size();
18     }
19     vis[currt] = 1;
20     h.push_back(alt);

```

```

21     nodo.push_back(currt);
22
23     for (int hijo : gf[currt])
24     {
25         if (!vis[hijo])
26         {
27             dfs(hijo, alt + 1);
28             h.push_back(alt);
29             nodo.push_back(currt);
30         }
31     }
32 }
33
34 int main()
35 {
36     vis.assign(n, 0);
37     primera.assign(n, -1);
38     gf.assign(n, vector<int>());
39     // st.clear();
40     h.clear();
41     nodo.clear();
42     //0i
43     dfs(0, 0);
44
45     //hacer un segmet tree sobre h que es la altura,
46     //y responder cout<<nodo[res] donde res es el
47     //indice del minimo en el rango, para minimo en
48     //rango, donde l y r son los dos nodos
49
50     return 0;
51 }

```

QueueMin

```

1 struct quemin
2 {
3     stack<pair<int,int>> bo, to;
4     void push(int n)
5     {
6         if(bo.empty())
7             bo.push(mp(n, n));
8         else
9             bo.push(mp(n, min(bo.top().s, n)));
10    }

```

```

11 void pop()
12 {
13     if(to.empty())
14     {
15         while(!bo.empty())
16         {
17             if(to.empty())
18                 to.push(mp(bo.top().f, bo.top().f));
19             else
20                 to.push(mp(bo.top().f, min(bo.top().f, to.
21                     top().s)));
22             bo.pop();
23         }
24         to.pop();
25     }
26     int mini()
27     {
28         int mini = MOD;
29         if(!bo.empty())
30             mini = bo.top().s;
31         if(!to.empty())
32             mini = min(mini, to.top().s);
33         return mini;
34     }
35 };
36
37 struct quemin
38 {
39     pair<int,int> bo[100010], to[100010];
40     int boto = -1, toto = -1, ax;
41     void push(int n)
42     {
43         ax = boto + 1;
44         if(boto == -1)
45             bo[ax] = mp(n, n);
46         else
47             bo[ax] = mp(n, min(bo[boto].s, n));
48         boto++;
49     }
50     void pop()
51     {
52         if(toto == -1)
53         {

```

```

54         while(boto > -1)
55         {
56             ax = boto + 1;
57             if(toto == -1)
58                 to[ax] = mp(bo[boto].f, bo[boto].f);
59             else
60                 to[ax] = mp(bo[boto].f, min(bo[boto].f, to
61                     [toto].s));
62             toto++;
63             boto--;
64         }
65         if(toto > -1)
66             toto--;
67     }
68     int mini()
69     {
70         int mini = MOD;
71         if(boto > -1)
72             mini = bo[boto].s;
73         if(toto > -1)
74             mini = min(mini, to[toto].s);
75         return mini;
76     }
77 };

```

SegmentTree

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4 typedef long long ll;
5
6 struct Data
7 {
8     ll cant = 0;
9     Data() { cant = 1e18; }
10
11     Data(ll c) { cant = c; }
12 };
13
14 struct SegTree
15 {

```

```

16 private:
17     vector<Data> st;
18
19 public:
20     int sz;
21     Data merge(Data a, Data b)
22     {
23         return min(a.cant, b.cant);
24     }
25     void init(int n, vector<Data> v)
26     {
27         while (__builtin_popcount(n) != 1)
28         {
29             n++;
30         }
31         st.resize(2 * n, Data());
32         sz = n; // solo n, NO 2*n
33         for (int i = 0; i < (int)v.size(); i++)
34         {
35             st[n + i] = v[i];
36         }
37
38         for (int i = n - 1; i > 0; --i)
39         {
40             st[i] = merge(st[i << 1], st[(i << 1) +
41                             1]);
42         }
43
44     void updateTreeNode(int p, Data nuevoValor) // 0
45         i pos
46     {
47         st[p + sz] = nuevoValor;
48         p = p + sz;
49         for (int i = p; i > 1; i >>= 1)
50             st[i >> 1] = merge(st[i], st[i ^ 1]);
51     }
52
53     Data query(int nodo, int left_nodo, int
54               right_nodo, int l_q, int r_q) //
55     {
56         // query(1,0,st.sz-1,l_q,r_q) tipo->[l_q,r_q
57         ]0i
58         /*

```

```

56         los indices de los nodos empieza desde 1;
57         */
58         if (l_q <= left_nodo && right_nodo <= r_q)
59         {
60             return st[nodo];
61         }
62         if (l_q > right_nodo || left_nodo > r_q)
63         {
64             return Data();
65         }
66
67         int mitad = (left_nodo + right_nodo) / 2; //
68         //[l:r] --> [l:mitad] [mitad+1:r]
69         return merge(query(nodo * 2, left_nodo,
70               mitad, l_q, r_q), query(nodo * 2 + 1,
71               mitad + 1, right_nodo, l_q, r_q));
72     }
73 };

```

SparseTable

```

1 struct SparseTable {
2     int n;
3     vector<int> log2;
4     vector<vector<int>> st;
5
6     SparseTable(const vector<int>& a) {
7         n = a.size();
8         log2.resize(n + 1);
9         log2[1] = 0;
10        for (int i = 2; i <= n; i++)
11            log2[i] = log2[i/2] + 1;
12
13        int k = log2[n] + 1;
14        st.assign(n, vector<int>(k));
15
16        for (int i = 0; i < n; i++) st[i][0] = a[i];
17
18        for (int j = 1; j < k; j++)
19            for (int i = 0; i + (1 << j) <= n; i++)
20                st[i][j] = min(st[i][j-1], st[i + (1
21                << (j-1))][j-1]);

```

```

22 // Consulta de minimo en el rango [l, r]oi
23 int query(int l, int r) {
24     int j = log2[r - l + 1];
25     return min(st[l][j], st[r - (1 << j) + 1][j
26         ]);
27 }
28 };

```

UnionFind

```

1 struct unionFind {
2     vi p;
3     unionFind(int n) : p(n, -1) {}
4     int findParent(int v) {
5         if (p[v] == -1) return v;
6         return p[v] = findParent(p[v]);
7     }
8     bool join(int a, int b) {
9         a = findParent(a);
10        b = findParent(b);
11        if (a == b) return false;
12        p[a] = b;
13        return true;
14    }
15 };

```

fenwickTree

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4 typedef long long ll;
5
6
7 struct Bit
8 {
9     private:
10        vector<ll> bit;
11        int sz;
12    public:
13

```

```

14 Bit(int n){bit.resize(n+1,0);this->sz=n;}
15
16 void init(vector<ll> vec)
17 {
18     for(int i=0;i<sz;i++)
19     {
20         update(i+1,vec[i]);
21     }
22 }
23
24 void update(int p,ll val)//1i
25 {
26     while(p<=sz)
27     {
28         bit[p]+=val;
29         p+=(p&(-p));
30     }
31 }
32 ll query(int p)//[1,p]1i
33 {
34     ll ans=0;
35     while(p>0)
36     {
37         ans+=bit[p];
38         p-=(p&-p);
39     }
40     return ans;
41 }
42
43 ll rquery(ll left,ll right)//[1,r]1i
44 {
45     return query(right)-query(left-1);
46 }
47 };

```

Strings

KMP

```

1 vector<int> prefix_function(string s)
2 {
3     int n = (int)s.length();
4     vector<int> pi(n);

```

```

5   for (int i = 1; i < n; i++)
6   {
7       int j = pi[i - 1];
8       while (j > 0 && s[i] != s[j])
9           j = pi[j - 1];
10      if (s[i] == s[j])
11          j++;
12      pi[i] = j;
13  }
14  return pi;
15  }
16
17  inline void solve()
18  {
19      string texto, sub;
20      cin >> texto >> sub;
21      int n = texto.size();
22      int m = sub.size();
23      vector<int> vec = prefix_function(sub + "$" +
24                                     texto);
25      cout << count(vec.begin(), vec.end(), m) << endl;
26  }

```

hashing

```

1  struct StrHash
2  { // Hash polinomial con exponentes decrecientes.
3      static constexpr ll ms[] = {1'000'000'007, 1'000
4                                     '000'403};
5      static constexpr ll b = 500'000'000;
6      vector<ll> hs[2], bs[2];
7      StrHash(string const &s)
8      {
9          int n = s.length();
10         for(int k=0;k<2;k++)
11         {
12             hs[k].resize(n + 1), bs[k].resize(n + 1,
13             1);
14             for(int i=0;i<n;i++)
15             {
16                 hs[k][i + 1] = (hs[k][i] * b + s[i])
17                                     % ms[k];

```

```

15         bs[k][i + 1] = bs[k][i] * b % ms[k];
16     }
17 }
18
19 ll get(int idx, int len) const
20 { // Hashes en 's[idx, idx+len)'
21     ll h[2];
22     for(int k=0;k<2;k++)
23     {
24         h[k] = hs[k][idx + len] - hs[k][idx] *
25             bs[k][len] % ms[k];
26         if (h[k] < 0)
27             h[k] += ms[k];
28     }
29     return (h[0] << 32) | h[1];
30 }
31
32 //concat substrings(or strings) from non necessary
33 //two differents strings [idx,idx+lex)
34 ll concat_cross_hashes(const StrHash& A, int i1, int
35 len1, const StrHash& B, int i2, int len2)
36 {
37     ll res[2];
38     for (int k = 0; k < 2; ++k)
39     {
40         // hash de substring A[i1..i1+len1-1]
41         ll h1 = A.hs[k][i1 + len1] - A.hs[k][i1] * A
42             .bs[k][len1] % A.ms[k];
43         if (h1 < 0) h1 += A.ms[k];
44
45         // hash de substring B[i2..i2+len2-1]
46         ll h2 = B.hs[k][i2 + len2] - B.hs[k][i2] * B
47             .bs[k][len2] % B.ms[k];
48         if (h2 < 0) h2 += B.ms[k];
49
50         // combinacion: h1 * b^len2 + h2
51         res[k] = (h1 * A.bs[k][len2] + h2) % A.ms[k];
52     }
53     return (res[0] << 32) | res[1];
54 }

```

```

53 //0 indexed
54 ll h=StrHash("Hola").get(0,0+"Hola".size());

```

hashing2d

```

1 struct Hashing
2 {
3     vector<vector<int>> hs;
4     vector<int> PWX, PWY;
5     int n, m;
6     static const int PX = 3731, PY = 2999, mod =
7         998244353;
8     Hashing() {}
9     Hashing(vector<string> &s)
10    {
11        n = (int)s.size(), m = (int)s[0].size();
12        hs.assign(n + 1, vector<int>(m + 1, 0));
13        PWX.assign(n + 1, 1);
14        PWY.assign(m + 1, 1);
15        for (int i = 0; i < n; i++)
16            PWX[i + 1] = 1LL * PWX[i] * PX % mod;
17        for (int i = 0; i < m; i++)
18            PWY[i + 1] = 1LL * PWY[i] * PY % mod;
19        for (int i = 0; i < n; i++)
20        {
21            for (int j = 0; j < m; j++)
22            {
23                hs[i + 1][j + 1] = s[i][j] - 'a' +
24                    1;
25            }
26        }
27        for (int i = 0; i <= n; i++)
28        {
29            for (int j = 0; j < m; j++)
30            {
31                hs[i][j + 1] = (hs[i][j + 1] + 1LL *
32                    hs[i][j] * PY % mod) % mod;
33            }
34        }
35        for (int i = 0; i < n; i++)
36        {
37            for (int j = 0; j <= m; j++)
38            {

```

```

36                hs[i + 1][j] = (hs[i + 1][j] + 1LL *
37                    hs[i][j] * PX % mod) % mod;
38            }
39        }
40    }
41    int get_hash(int x1, int y1, int x2, int y2)
42    { // 1-indexed
43        assert(1 <= x1 && x1 <= x2 && x2 <= n);
44        assert(1 <= y1 && y1 <= y2 && y2 <= m);
45        x1--;
46        y1--;
47        int dx = x2 - x1, dy = y2 - y1;
48        return (1LL * (hs[x2][y2] - 1LL * hs[x2][y1]
49            * PWY[dy] % mod + mod) % mod -
50            1LL * (hs[x1][y2] - 1LL * hs[x1][y1]
51                * PWY[dy] % mod + mod) % mod *
52            PWX[dx] % mod + mod) %
53            mod;
54    }
55    int get_hash()
56    {
57        return get_hash(1, 1, n, m);
58    }
59 };

```

Geometry

Point

```

1 /*typedef double T;
2 typedef complex<T> pt;
3 #define x real()
4 #define y imag()*/
5
6 //typedef long long ll;
7 //typedef long double ll;
8
9 struct point
10 {
11     ll x, y;
12     point() {}
13     point(ll x, ll y): x(x), y(y) {}

```

```

14 point operator -(point p) {return point(x - p.x, y
    - p.y);}
15 point operator +(point p) {return point(x + p.x, y
    + p.y);}
16 ll sq() {return x * x + y * y;}
17 double abs() {return sqrt(sq());}
18 ll operator ^(point p) {return x * p.y - y * p.x;}
19 ll operator *(point p) {return x * p.x + y * p.y
    ;}
20 point operator *(ll a) {return point(x * a, y *
    a);}
21 bool operator <(const point& p) const {return x ==
    p.x ? y < p.y : x < p.x;}
22 bool left(point a, point b) {return ((b - a) ^ (*
    this - a)) >= 0;}
23 ostream& operator<<(ostream& os) {
24     return os << "(" << x << ", " << y << ")";
25 }
26
27 };
28
29 void polarSort(vector<point>& v) {
30     sort(v.begin(), v.end(), [] (point a, point b) {
31         const point origin{0, 0};
32         bool ba = a < origin, bb = b < origin;
33         if (ba != bb) { return ba < bb; }
34         return (a^b) > 0;
35     });
36 }

```

convexHull

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 typedef long long ll;
5
6 struct pt
7 {
8     double x, y;
9     bool operator==(pt const &t) const
10     {
11         return x == t.x && y == t.y;

```

```

12     }
13 };
14
15 int orientation(pt a, pt b, pt c)
16 {
17     double v = a.x * (b.y - c.y) + b.x * (c.y - a.y) +
        c.x * (a.y - b.y);
18     if (v < 0)
19         return -1; // clockwise
20     if (v > 0)
21         return +1; // counter-clockwise
22     return 0;
23 }
24
25 bool cw(pt a, pt b, pt c, bool include_collinear)
26 {
27     int o = orientation(a, b, c);
28     return o < 0 || (include_collinear && o == 0);
29 }
30 bool collinear(pt a, pt b, pt c) { return
    orientation(a, b, c) == 0; }
31
32 void convex_hull(vector<pt> &a, bool
    include_collinear = false)
33 {
34     pt p0 = *min_element(a.begin(), a.end(), [](pt a,
        pt b)
35         { return make_pair(a.y, a.x) <
            make_pair(b.y, b.x); });
36     sort(a.begin(), a.end(), [&p0](const pt &a, const
        pt &b)
37     {
38         int o = orientation(p0, a, b);
39         if (o == 0)
40             return (p0.x-a.x)*(p0.x-a.x) + (p0.y-a.y)
                *(p0.y-a.y)
41                 < (p0.x-b.x)*(p0.x-b.x) + (p0.y-b.y)
                    *(p0.y-b.y);
42         return o < 0; });
43     if (include_collinear)
44     {
45         int i = (int)a.size() - 1;
46         while (i >= 0 && collinear(p0, a[i], a.back()))
47             i--;

```

```

48     reverse(a.begin() + i + 1, a.end());
49 }
50
51 vector<pt> st;
52 for (int i = 0; i < (int)a.size(); i++)
53 {
54     while (st.size() > 1 && !cw(st[st.size() - 2],
55         st.back(), a[i], include_collinear))
56         st.pop_back();
57     st.push_back(a[i]);
58 }
59
60 if (include_collinear == false && st.size() == 2
61     && st[0] == st[1])
62     st.pop_back();
63
64 a = st;
65 }
66
67 vector<pt> a;
68
69 int main()
70 {
71     int n; cin >> n;
72     a.resize(n);
73     // leer como puntos
74     convex_hull(a, true); // true incluye colinear, false
75                             no lo hace
76 }

```

Primalidad

RabinMiller

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5
6 using u64 = uint64_t;
7 using u128 = __uint128_t;

```

```

8
9 u64 binpower(u64 base, u64 e, u64 mod) {
10     u64 result = 1;
11     base %= mod;
12     while (e) {
13         if (e & 1)
14             result = (u128)result * base % mod;
15         base = (u128)base * base % mod;
16         e >>= 1;
17     }
18     return result;
19 }
20
21 bool check_composite(u64 n, u64 a, u64 d, int s) {
22     u64 x = binpower(a, d, n);
23     if (x == 1 || x == n - 1)
24         return false;
25     for (int r = 1; r < s; r++) {
26         x = (u128)x * x % n;
27         if (x == n - 1)
28             return false;
29     }
30     return true;
31 };
32
33 bool MillerRabin(u64 n, int iter=5) { // returns
34     true if n is probably prime, else returns false.
35     cout << iter << endl;
36     if (n < 4)
37         return n == 2 || n == 3;
38
39     int s = 0;
40     u64 d = n - 1;
41     while ((d & 1) == 0) {
42         d >>= 1;
43         s++;
44     }
45
46     for (int i = 0; i < iter; i++) {
47         int a = 2 + rand() % (n - 3);
48         if (check_composite(n, a, d, s))
49             return false;
50     }
51     return true;

```

```

51 }
52
53
54 int main()
55 {
56
57     cout<<MillerRabin(1000000001,30)<<endl;
58     return 0;
59 }

```

pollardRho

```

1  #include <iostream>
2  #include <cstdlib>
3  #include <cstdio>
4  #include <cmath>
5  #include <cassert>
6  #include <map>
7
8  using namespace std;
9
10 typedef long long ll;
11
12 #define forn(i, n) for (int i = 0; i < (int)(n); i++)
13 #define forsn(i, s, n) for (int i = int(s); i < (int)(n); i++)
14
15 // rabin miller
16
17 ll potlog(ll a, ll b, const ll M)
18 {
19     ll res = 1;
20     while (b)
21     {
22         if (b % 2)
23             res = (__int128(res) * a) % M;
24         a = (__int128(a) * a) % M;
25         b /= 2;
26     }
27     return res;
28 }
29

```

```

30 bool primo(ll n)
31 {
32     if (n < 2)
33         return false;
34     if (n == 2)
35         return true;
36     ll D = n - 1, S = 0;
37     while (D % 2 == 0)
38     {
39         D /= 2;
40         S++;
41     }
42     // n-1 = 2^S * D
43     static const int STEPS = 16;
44     forn(pasos, STEPS)
45     {
46         const ll A = 1 + rand() % (n - 1);
47         ll M = potlog(A, D, n);
48         if (M == 1 || M == (n - 1))
49             goto next;
50         forn(k, S - 1)
51         {
52             M = (__int128(M) * M) % n;
53             if (M == (n - 1))
54                 goto next;
55         }
56         return false;
57     next:;
58     }
59     return true;
60 }
61
62 // pollard's rho
63
64 ll mcd(ll a, ll b) { return (a == 0) ? b : mcd(b % a, a); }
65
66 ll factor(ll n)
67 {
68     static ll A, B;
69     A = 1 + rand() % (n - 1);
70     B = 1 + rand() % (n - 1);
71     #define f(x) ((__int128(x) * (x + B)) % n + A)
72     ll x = 2, y = 2, d = 1;

```

```

73     while (d == 1 || d == -1)
74     {
75         x = f(x);
76         y = f(f(y));
77         d = mcd(x - y, n);
78     }
79     return abs(d);
80 }
81
82 map<ll, ll> fact;
83
84 void factorize(ll n)
85 {
86     assert(n > 0);
87     while (n > 1 && !primo(n))
88     {
89         ll f;
90         do
91         {
92             f = factor(n);
93         } while (f == n);
94         n /= f;
95         factorize(f);
96         for (auto &it : fact)
97             while (n % it.first == 0)
98             {
99                 n /= it.first;
100                it.second++;
101            }
102     }
103     if (n > 1)
104         fact[n]++;
105 }
106
107 int main()
108 {
109     ll N;
110     while (cin >> N && N)
111     {
112         fact.clear();
113         factorize(N);
114         for (const auto &it : fact)
115             cout << it.first << "^" << it.second <<
116                 " ";

```

```

116         cout << endl;
117     }
118     return 0;
119 }

```

Graphs

isDag

```

1 vector<vector<int>> gf; // lista de adyacencia
2 vector<int> visited; // 0 = no visitado, 1 =
3     visitando, 2 = visitado
4
5 bool dfs(int u) {
6     visited[u] = 1; // visitando
7     for (int v : gf[u]) {
8         if (visited[v] == 1) return true; // ciclo
9         detectado
10        if (visited[v] == 0 && dfs(v)) return true;
11    }
12    visited[u] = 2; // visitado
13    return false;
14 }
15
16 bool isDAG(int n) {
17     visited.assign(n, 0);
18     for (int i = 0; i < n; i++)
19         if (visited[i] == 0 && dfs(i))
20             return false; // hay ciclo
21     return true; // no hay ciclos
22 }

```

isDagv2

```

1 vector<vector<int>> gf;
2 vector<bool> vis;
3 set<int> st;
4 bool sw = 1;
5 int n;
6
7 void isDAG(int nodo)
8 {

```

```

9      // no olvidar recorrer con un for todo
10     //ya que no siempre estan conectados
11     if (!sw) return;
12     vis[nodo] = 1;
13     st.insert(nodo);
14     for (auto hijo : gf[nodo])
15     {
16         if (st.count(hijo) == 1)
17         {
18             sw = 0;
19             return;
20         }
21         if (!vis[hijo])
22         {
23             vis[hijo] = 1;
24             isDAG(hijo);
25         }
26     }
27     st.erase(nodo);
28 }

```

toposort

```

1  #include <bits/stdc++.h>
2  using namespace std;
3  int n; // number of vertices
4  vector<vector<int>> gf;
5  vector<bool> vis;
6  vector<int> ans;
7
8  void dfs(int v) { // 0i
9      vis[v] = true;
10     for (int u : gf[v]) {
11         if (!vis[u]) {
12             dfs(u);
13         }
14     }
15     ans.push_back(v);
16 }
17
18 void topological_sort() {
19     vis.assign(n, false);
20     ans.clear();

```

```

21     for (int i = 0; i < n; ++i) {
22         if (!vis[i]) {
23             dfs(i);
24         }
25     }
26     reverse(ans.begin(), ans.end());
27 }

```

Overloads

pq

```

1 struct cmp
2 {
3     //mayor tiene prioridad
4     bool operator()(const int& a, const int& b)
5     const {
6         return a < b;
7     }
8 };
9 //priority_queue<Data, vector<Data>, cmp> pq

```

set

```

1 struct cmpST
2 {
3     //de menor a mayor
4     bool operator()(const int &a, const int &b)
5     {
6         return a < b;
7     }
8 };
9
10 //set<int, cmpST> st;

```